

HBSC: Modeling

Updated: 2026-08-14

Data Overview

This data is prepared in hbsc.rmd file. It's dimensions are:

```
## Rows: 244097 Columns: 17
## -- Column specification -----
## Delimiter: ","
## chr (6): country_name, pmsu_lmh, family_support_hml, friends_support_hml, I...
## dbl (11): seqno_int, lifesat_low, mental_complaintsY, pmsu_yn, family_suppor...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

## [1] 244097      17
```

Univariate Exploration and Preparation

The list of vars and it's type is:

```
## Rows: 244,097
## Columns: 17
## $ seqno_int      <dbl> 100001, 100002, 100004, 100005, 100007, 100008, 1~
## $ country_name   <chr> "Albania", "Albania", "Albania", "Albania", "Alba~
## $ lifesat_low      <dbl> 0, 0, 0, NA, 0, 0, NA, 1, NA, NA, 0, 0, 0, 0, 0, ~
## $ mental_complaintsY <dbl> 1, 0, 0, 0, 1, 1, 0, NA, NA, 0, 0, NA, 0, 0, NA, ~
## $ pmsu_lmh        <chr> "Med", "Med", "High", "Med", "Med", "Low", "Med",~
## $ pmsu_yn         <dbl> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, NA, 0, NA, 0, 0, NA, 0~
## $ family_support_high <dbl> 1, 1, 1, 1, 0, 1, 1, NA, 1, 1, NA, NA, 1, 1, 1, 1~
## $ family_support_hml <chr> "High", "High", "High", "High", "Low", "High", "H~
## $ friends_support_high <dbl> 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, NA, 0, 1, 1, 1, 0, ~
## $ friends_support_hml <chr> "High", "High", "Med", "High", "High", "High", "H~
## $ IRRELFAS_LMH_r   <chr> "Med60", NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ talkfatherYes    <dbl> 1, 1, 1, 1, 0, 0, 1, NA, NA, 1, NA, NA, 1, NA, 1,~
## $ talkmotherYes    <dbl> 1, 1, 1, 1, 1, 1, 1, NA, NA, 1, NA, NA, 1, NA, NA~
## $ beenbulliedYes   <dbl> 0, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0~
## $ cbeenbulliedYes  <dbl> 1, 1, 1, 0, 0, 0, 0, 0, NA, 0, 0, 0, 0, NA, 0, 0,~
## $ age_r           <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ sex_r           <chr> "Boy", "Boy", "Boy", "Boy", "Boy", "Boy", "Boy", ~
```

We now look one by one and will:

1. Coerce to correct var type for modeling (mostly factors)
2. Note potential of prediction

Coerce first

For modeling, many vars need to be coerced so I start there before univ EDA.

```
d <- d %>%
  mutate(lifesat_low = factor(lifesat_low, levels = c("0", "1")),
         mental_complaintsY = factor(mental_complaintsY, levels = c("0", "1")),
         pmsu_lmh = factor(pmsu_lmh, levels = c("low", "med", "high")),
         pmsu_yn = factor(pmsu_yn, levels = c("0", "1")),
         family_support_high = factor(family_support_high, levels = c("0", "1")),
         family_support_hml = factor(family_support_hml, levels = c("low", "med", "high")),
         friends_support_high = factor(friends_support_high, levels = c("0", "1")),
         friends_support_hml = factor(friends_support_hml, levels = c("low", "med", "high")),
         IRRELFAS_LMH_r = factor(IRRELFAS_LMH_r, levels = c("Low20", "Med60", "High20")),
         talkfatherYes = factor(talkfatherYes, levels = c("0", "1")),
         talkmotherYes = factor(talkmotherYes, levels = c("0", "1")),
         beenbulliedYes = factor(beenbulliedYes, levels = c("0", "1")),
         cbeenbulliedYes = factor(cbeenbulliedYes, levels = c("0", "1")),
         age_r = factor(age_r, levels = c("11", "13", "15")),
         sex_r = factor(sex_r)
  )
```

seqno_int

Children id and confirmed it's unique.

```
length(unique(d$seqno_int)) == nrow(d)
```

```
## [1] TRUE
```

\$ lifesat_low 13% report having low satisfaction and still ~2% of NAs that need to be fixed.

```
## # A tibble: 3 x 3
##   lifesat_low      n    pct
##   <fct>         <int> <dbl>
## 1 0           207789 0.851
## 2 1           31796 0.13
## 3 <NA>         4512 0.018
```

mental_complaintsY 26% report mental health issues with ~5% of NAs that need to be fixed.

```
## # A tibble: 3 x 3
##   mental_complaintsY      n    pct
##   <fct>                 <int> <dbl>
## 1 0           167082 0.684
## 2 1           64133 0.263
## 3 <NA>         12882 0.053
```



```

####$ friends_support_high 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, NA, 0, 1, 1, 1, 0, 1, NA, 1, 0, 1, 1, 0, 0, 0, 1, NA...
####$ friends_support_hml "high", "high", "med", "high", "high", "high", "high", "high", "high", "high", "high",
NA,...
####$ IRRELFAS_LMH_r "Med60", NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "Med60", NA,
"Med60", "Med60"...
####$ talkfatherYes 1, 1, 1, 1, 0, 0, 1, NA, NA, 1, NA, NA, 1, NA, 1, 1, 1, NA, 1, 0, 1, 1, 1, 1, 1, 1...
####$ talkmotherYes 1, 1, 1, 1, 1, 1, 1, NA, NA, 1, NA, NA, 1, NA, NA, 1, 1, NA, NA, 0, 1, 1, 1, 1, 1,...
####$ beenbulliedYes 0, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0...
####$ cbeenbulliedYes 1, 1, 1, 0, 0, 0, 0, 0, NA, 0, 0, 0, 0, NA, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0...
####$ age_r NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
####$ sex_r

```

for later: keep complete obs only model like a rusian doll